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Large language models (LLMs) are increasingly deployed as interacting agents that exchange information, negotiate decisions, and coordinate their actions over multiple turns. Such multi-agent systems (MAS) promise to combine information and capabilities distributed across multiple agents, yet recent work has exposed substantial failures that arise specifically through interaction. Agents fail to proactively acquire information that is available only to others (Li et al., 2026a), communicate actively without converting those exchanges into effective distributed reasoning (Zhang et al., 2026b), and remain unable to coordinate effectively even when relevant information is made transparent (Yao et al., 2026). Broader analyses similarly identify inter-agent failures as a distinct source of error beyond mistakes made by individual agents in isolation (Cemri et al., 2025; Anthropic, 2026). These findings raise a basic question: what capabilities are missing when individually capable agents fail to interact effectively?

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Classical models of interactive decision making provide a principled way to make this question more precise. In an interactive POMDP, an agent maintains uncertainty not only over the physical environment, but also over models of other agents, and plans its own actions with respect to this interactive state (Gmytrasiewicz & Doshi, 2005). Related work on opponent modeling similarly couples inference about another agent with planning a response to the inferred model (Huang et al., 2024). Viewed through this lens, effective interaction requires at least two computations: **interactive-state inference**, which construct belief of other agents relevant information, and **interaction planning**, which uses that belief to evaluate what to do next. Importantly, these computations are coupled over time. An inferred state changes the value of an interaction action; the selected interaction then changes how other agents respond and therefore what evidence becomes available for subsequent inference. Established models of multi-agent decision making identify these two computations as fundamental components of interaction, motivating us to ask whether current LLM agents can reliably perform them together (Riemer et al., 2025).

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How can these capabilities be diagnosed and trained in a controlled way? Games provide a natural substrate because their information structure, utilities, legal actions, and strategic consequences can be specified and verified exactly. Recent work has accordingly used multi-agent games and self-play to train strategic or transferable reasoning in LLMs (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b). However, these environments are primarily designed around game-level performance rather than separate supervision of interactive-state inference and the planning that follows from it. Compact self-play games make scalable training possible but often abstract away the combination of latent information and heterogeneous incentives, whereas richer social games can entangle deduction, communication, deception, commitment and execution within the same trajectory (Light et al., 2023). To obtain precise supervision for the interaction structure above, we instead extend an existing commitment-based multi-party negotiation game (Benac et al., 2026), which makes the environment becomes incomplete information and mixed incentives. Incomplete information makes relevant properties of other agents latent and therefore creates an inference problem, while mixed incentives make those inferred properties consequential for planning because the value of an interaction depends on what the other agents know, prefer, or are likely to do. Their combination captures an important class of real interactions in which agents act on behalf of different principals—for example, coordinating around private scheduling constraints (Zou et al., 2026),

negotiating under asymmetric private objectives (O’Neill et al., 2026), or interacting repeatedly on behalf of distinct economic interests (Li et al., 2026b).

Using this controlled setting, we separately diagnose failures in interactive-state inference and interaction planning, then construct strategic game slices that provide oracle supervision for both. We train on these slices and evaluate whether the resulting improvements generalize to unseen strategic configurations and transfer to external multi-agent environments.

Our contributions are threefold:

- **A controlled diagnosis of interactive reasoning in LLM-based MAS.** Grounded in classical interactive decision making, we study interactive-state inference and interaction planning in incomplete-information, mixed-incentive coordination, and separately measure failures in the two while retaining their sequential dependence.
- **A compact game construction for verifiable social-reasoning supervision.** We show how private preferences, consequential commitments, and information-revealing actions can instantiate a practically relevant interactive-decision problem while providing exact latent-state targets and counterfactual action values.
- **Targeted game supervision and transfer.** We construct strategic game slices around diagnosed interaction decisions, use their oracle structure to supervise inference and planning, and test whether the resulting capabilities generalize to unseen strategic configurations and external multi-agent environments.

2 RELATED WORKS

Classical multi-agent decision making couples inference about other agents with downstream action selection, from opponent models and interactive beliefs (Carmel & Markovitch, 1996; Gmytrasiewicz & Doshi, 2005; Albrecht & Ramamoorthy, 2013) to belief-space coordination, inference from actions, and opponent-aware planning (Macindoe et al., 2012; Wu et al., 2021; Foerster et al., 2019; Huang et al., 2024). For LLM agents, explicit mental-state or behavioral representations support collaboration, adaptation, trust tracking, and planning (Li et al., 2023; Cross et al., 2025; Jha et al., 2026; Mu et al., 2026; Zhou et al., 2026; Feng et al., 2026; Goel et al., 2026; Hwang et al., 2026). However, accurate mental-state prediction need not yield useful behavior (Riemer et al., 2025; Sobotka et al., 2026).

Recent MAS benchmarks expose strategic failures under private information, mixed incentives, and imperfect communication. Commitment-based negotiation, self-negotiated contracts, and multi-party conquest model consequential agreements and coalition dynamics (Benac et al., 2026; Wyse et al., 2026; O’Neill et al., 2026); CalBench, SOTOPIA-TOM, and SNEAK make information disclosure and privacy strategically consequential (Zou et al., 2026; YS et al., 2026; Avsian & Heck, 2026). Broader evaluations cover collaboration, controlled interaction, long-horizon commerce, and communication barriers (Zhu et al., 2025; Chen et al., 2026; Li et al., 2026b; Xuan et al., 2026). Diagnostic work further finds failures in distributed information pooling, distributed computation, dynamic grounding, and belief-sensitive negotiation (Li et al., 2026a; Zhang et al., 2026b; Yao et al., 2026; Zhang et al., 2026a; Agashe et al., 2025); these complement broad failure taxonomies and accounts of long-lived peer interaction (Cemri et al., 2025; Anthropic, 2026).

Game-based training has been used to induce transferable strategic and social reasoning through self-play, multi-game reinforcement learning, and social reasoning supervision (Yuan et al., 2026; Liu et al., 2026; Lyu et al., 2026; He et al., 2026b; Hua et al., 2026; He et al., 2026a; Wu et al., 2026). Cross-environment cooperation and Other-Play study generalization across tasks and partners (Jha et al., 2025; Hu et al., 2020). Curriculum work selects terminal-proximal or high-learning-value subgames (West et al., 2020; Chen et al., 2024), while LLM curricula can also induce undesirable strategic abstractions (Madmoun & Lahlou, 2026). Similarity-based inference has also been proposed as a route to cooperation among foundation-model agents (Meulemans et al., 2026).

3 PRELIMINARIES

3.1 FROM INTERACTIVE UNCERTAINTY TO TWO CAPABILITIES

Coordination with private information is naturally an interactive partially observable decision problem. For agent i , an I-POMDP is $\mathcal{I}_i = \langle IS_i, A, T_i, \Omega_i, O_i, R_i \rangle$, $IS_i = S \times \prod_{j \neq i} M_j$ where S is the physical state, M_j models agent j , A is the joint action space, and T_i, O_i, R_i specify transition, observation, and reward (Gmytrasiewicz & Doshi, 2005). Uncertainty therefore covers not only the world but also others’ information, preferences, and policies. This view underlies opponent modelling and belief-space coordination (Carmel & Markovitch, 1996; Huang et al., 2024; Goel et al., 2026); the observation–belief–action links nevertheless remain fragile for LLM agents (Sobotka et al., 2026; Jha et al., 2026). Let $b_t^i \in \Delta(IS_i)$ be agent i ’s belief after public history h_t . An observation updates it as $b_{t+1}^i = \tau_i(b_t^i, a_t^i, o_{t+1}^i)$, while an action is selected by belief-conditioned planning, $a_t^{i,*} \in \arg \max_{a \in A_i} \mathbb{E} \left[\sum_{k=t}^T \gamma^{k-t} R_i(x_k, a_k) \mid b_t^i, a_t^i = a, \pi_{-i} \right]$. We call these computations *partner inference* (B), which maintains the part of b_t^i predictive of others, and *partner-conditioned planning* (P), which evaluates actions under that belief. They form the loop $B_t \rightarrow P_t \rightarrow a_t^i \rightarrow o_{t+1}^i \rightarrow B_{t+1}$: belief changes the best interaction, while the chosen interaction determines the next evidence. This is a functional decomposition, not a claim about separate neural modules.

3.2 BENAC-I: NEGOTIATION WITH PRIVATE PREFERENCES

To expose both computations while retaining exact supervision, we extend the commitment-based negotiation game BENAC (Benac et al., 2026). We call the resulting incomplete-information variant BENAC-I. An instance is

$$\mathcal{E} = \langle \mathcal{N}, \mathbf{K}, \mathcal{G}, \Gamma, \delta, \rho, C_0, \mathcal{L}, \{\Theta_i\}_{i \in \mathcal{N}}, \mu_0, \mathbf{q} \rangle. \quad (1)$$

Here $\mathcal{N}$ is the player set, K_i counts i ’s binary commitment coordinates, and each public goal $g \in \mathcal{G}$ has requirements $\Gamma_g \subseteq \{(i, k) : k < K_i\}$ and mode $\delta_g \in \{\text{binary}, \text{linear}\}$. The proposer schedule ρ and legal-action relation $\mathcal{L}$ govern play from the initially empty, irreversible commitment state C_0 . Each player privately observes a preference type $v_i \in \Theta_i \subseteq \{-1, 0, +1\}^{|\mathcal{G}|}$, drawn from common prior μ_0 ; entries mean AVOID, NEUTRAL, or WANT. Vector $\mathbf{q}$ records remaining investigation budgets. For commitment state C , goal satisfaction and terminal utility are

$$S_g(C) = \begin{cases} \mathbf{1}[c_{ik} = 1, \forall (i, k) \in \Gamma_g], & \delta_g = \text{binary}, \\ |\Gamma_g|^{-1} \sum_{(i, k) \in \Gamma_g} c_{ik}, & \delta_g = \text{linear}, \end{cases} \quad U_i(C; v_i) = \sum_g v_{ig} S_g(C). \quad (2)$$

On a proposal turn, a player may PASS, OFFER new commitments, or use INVESTIGATE(j, g) when $q_i > 0$. Acceptance irreversibly binds both sides’ proposed additions; rejection binds none; choosing from a two-offer proposal binds only the selected option. Investigation instead consumes the turn and one unit of q_i , leaves C unchanged, and privately reveals v_{jg} ; the target, but not the answer, is public. Negotiation therefore supplies indirect evidence and investigation supplies costly direct evidence within the same action space.

Reference teacher and labels. The teacher is an information-limited policy profile, not an omniscient player. At an information set $I_j = (h, v_j, r_j)$ —public history, own preferences, and private investigation results—it chooses

$$\text{supp } \pi_j^*(\cdot \mid I_j) \subseteq \arg \max_{a \in \mathcal{L}(I_j)} \mathbb{E}_{v_{-j} \sim b_j(\cdot \mid I_j)} [U_j(C_T; v_j) \mid a, \pi_{-j}^*], \quad (3)$$

breaking own-utility ties by total utility to others and mixing over residual ties. We enumerate the finite game tree, initialize legal actions uniformly, and synchronously update information-set best responses until the whole contingent profile is stable. We retain it only after a unilateral-deviation check; cycles and resource-limit failures yield no label. This defines a reproducible reference, not a unique or general-sum Nash equilibrium.

The same profile supplies both labels. An observed action gives the exact finite-support update $b_{t+1}^i(v_{-i}) \propto b_t^i(v_{-i}) \pi_j^*(u_t^j \mid I_t^j, v_j)$, treating i ’s actions as interventions and conditioning on any

private investigation result; this is the B target. For P , exact continuation gives $Q_i^*(I_i, a)$ for every legal action and hence every optimal action. Both labels thus share one declared behavioral model and terminal objective.

4 DIAGNOSE

We next test whether an LLM can recover the partner state needed for a decision (B), choose an effective interaction when that state is supplied (P), and compose the two. Final game reward alone cannot distinguish these failures: a poor action may follow either a wrong judgment or poor use of a correct one, while two errors may occasionally cancel.

4.1 MATCHED SEMANTIC INTERVENTIONS

We construct 16 held-out two-player structures, evenly divided between binary and linear goals. Each structure yields a matched pair. In the *voluntary* case, the displayed partner actions are observations generated by the fixed reference policy; in the *preset* case, the same actions are imposed interventions and hence provide no evidence about preference. The two cases end at exactly the same physical state and have the same legal actions, per-world payoff table, and fixed continuation policy. Changing evidential status therefore changes only the posterior weights used to evaluate this otherwise identical planning problem. Selection uses teacher quantities only. In every pair, the correct semantic judgment changes. We retain a structure only if, on each side, every posterior compatible with that semantic judgment has the same exact optimal-action set, and the two invariant sets are disjoint. Thus the deployed judgment is sufficient for the decision while the inference difference remains decision-relevant.

The model-facing interface is the one used by our agent. For the single unknown partner-goal preference, B returns (i) the complete subset of WANT, NEUTRAL, and AVOID that remains possible and (ii) the uniquely most supported value, or UNDETERMINED. It never returns probabilities. We then query fresh planning contexts with either the model judgment $\hat{B}$ or the correct semantic judgment B^* . These controlled planning prompts retain the complete current decision state—the binding commitments, remaining turns, investigation quota, legal actions, and fixed continuation policy—but remove the behavioral chronology from which B was inferred. Before inserting the judgment, the voluntary and preset planning requests are byte-identical. Thus the supplied judgment is the only route by which their evidential distinction reaches P . A third condition asks for a native history-to-action decision without an explicit judgment. Exact posteriors and per-world payoffs remain in the backend and are used only for scoring. Importantly, B^* denotes the correct deployed semantic summary, not the full posterior. We certify its decision sufficiency by decomposing the compatible posterior region into convex polytopes and exactly enumerating their rational vertices. Since action values are affine in the posterior, an identical optimal-action set at every vertex proves invariance throughout the region. We conservatively include the closure of strict support and favored-margin constraints. In an auxiliary condition on eight voluntary structures, B additionally receives a qualitative reference-planner statement—e.g., that the observed sequence is impossible, more likely, or less likely under each preference—but still receives no numerical belief. This tests whether failures arise from modeling the partner’s plan or from subsequently integrating that behavioral evidence.

Let $F_c(a, \theta)$ be the certified payoff of legal action a in case c when the unknown preference is θ , under the same fixed continuation, and let b_c^* be the backend posterior. We score every action using

$$V_c(a) = \sum_{\theta} b_c^*(\theta) F_c(a, \theta), \quad \text{Reg}_c(a) = \max_{a'} V_c(a') - V_c(a). \quad (4)$$

Thus an action prompted by an incorrect judgment is still evaluated in the correct world distribution. We also measure the paired judgment-repair effect

$$\Delta_B = V_c(\hat{a}(B^*)) - V_c(\hat{a}(\hat{B})), \quad (5)$$

using deterministic decoding so that the two calls differ only in the supplied judgment. A positive value means that repairing B improves this planner; it is not an additive estimate of a standalone “B error.” Invalid tool outputs are missing rather than replaced with a default action.

Table 1: Native-semantic diagnosis on 16 matched structures (32 cases), all decision-sufficient. Regrets are structure means over each condition’s valid outputs; repair uses only identical doubly valid pairs.

Quantity	Old BP-99	New BP-99
Semantic B exact	15.6%	15.6%
Voluntary-case B exact	0/16	0/16
Possible-set exact	43.8%	50.0%
Correct- B P valid	25/32	29/32
Correct- B P regret	0.828	1.135
Model- B P valid	20/32	27/32
Model- B P regret	0.568	0.845
Native end-to-end valid	28/32	32/32
Native end-to-end regret	0.690	0.854
Paired cases (structures)	16 (11)	24 (14)
Action changed under repair	7/16	16/24
Repair negative / zero / positive	3/13/0	6/18/0
Mean paired repair Δ_B	-0.144	-0.310
Qualitative-plan B exact gain	0 pp	0 pp

4.2 INFERENCE FAILURE AND CLEAN PLANNING INTERVENTION

Table 1 compares two BP checkpoints on the same frozen suite. Exact B requires both the complete possible set and the correct favored field. All 408 calls per checkpoint completed, and the three replicas were identical; replicas are therefore collapsed rather than treated as independent samples.

Both checkpoints fail to recover the voluntary evidence: neither produces one exact judgment in 16 voluntary cases. Their five exact judgments occur only in the preset half. The new model’s possible-set gain also comes from preset cases (16/16 versus 14/16); both score 0/16 on voluntary possible sets. Qualitative descriptions of the reference plan change no exact judgment.

Planning remains defective after inference is fixed. With the correct, decision-sufficient B^* , regret is positive on 19/25 valid old-model cases and 27/29 new-model cases. Hence this failure cannot be attributed to missing posterior magnitudes. The new model produces valid actions more often, yet has higher regret. On the common valid subsets, correct- B regret is 0.802 versus 1.130 over 24 cases spanning all 16 structures, while end-to-end regret is 0.690 versus 0.887 over 28 cases and 14 structures. Thus improved protocol compliance does not translate into better decisions.

Repair exposes the coupling between the two capabilities. Changing only the semantic judgment changes the action in 7/16 and 16/24 paired cases, so the planner does respond to B ; however, no repair improves utility. Three old and six new cases become worse, while the rest are utility-equivalent. On the 14 cases (10 structures) valid for both checkpoints, mean repair is -0.158 and -0.550, respectively. We interpret this as a misaligned belief-to-action interface and error cancellation, not as evidence that the planner ignores B or as an additive decomposition of error.

Illustrative failure patterns. In `binary-20796:preset`, the new model conflates a goal’s required action with preference, calling Blair’s Library preference “publicly known” and favoring WANT; this wrong B nevertheless elicits the certified optimal offer. Replacing it with the correct UNDETERMINED judgment makes the same planner choose INVESTIGATE, despite stating that the game ends after this turn, increasing regret from 0 to 2. This is a recurrent opportunity-cost failure: under correct B , all 7 old-model and all 14 new-model investigations have positive regret. The trace makes the negative repair concrete: an inference error and a planning error can cancel.

5 EXPERIMENT

6 CONCLUSION

LIMITATION AND FUTURE WORK

AI USE STATEMENT

We used generative AI tools to assist with literature discovery, reference formatting, and drafting and editing portions of this manuscript. All AI-assisted text and bibliographic metadata were reviewed by the authors. The authors take responsibility for the final content of the paper, including its claims, experimental artifacts, and citations.

ETHICS STATEMENT

This work studies LLM-agent interaction in a synthetic, simulated negotiation environment and does not involve human participants or personal data. The environment includes mixed incentives and strategic information disclosure, which can surface behaviors such as manipulation or unhelpful information sharing. We use these behaviors only for controlled evaluation and training of coordination capabilities. We document the game rules, oracle policies, and limitations so that the resulting methods are not interpreted as deployment policies for real-world autonomous agents.

REPRODUCIBILITY STATEMENT

The main text and appendix specify the BENAC-P game rules, information structure, oracle belief filtering, and reference planning procedure. The appendix reports the diagnostic positions, prompts, evaluated model version, decoding configuration, controls, aggregation procedure, random seed, and provenance information used for the reported results. These details are intended to make the diagnostic suite and its measurements reproducible.

REFERENCES

- Saaket Agashe, Yue Fan, Anthony Reyna, and Xin Eric Wang. LLM-coordination: Evaluating and analyzing multi-agent coordination abilities in large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 8053–8072. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-naacl.448.
- Stefano V. Albrecht and Subramanian Ramamoorthy. A game-theoretic model and best-response learning method for ad hoc coordination in multiagent systems. In *Proceedings of the 2013 International Conference on Autonomous Agents and Multi-Agent Systems*, pp. 1155–1156, 2013.
- Anthropic. Patterns and problems in emerging multiagent systems. Anthropic Research, August 2026. Published August 13, 2026.
- Adar Avsian and Larry Heck. SNEAK: Evaluating strategic communication and information leakage in large language models. In *Conference on Language Modeling (COLM)*, 2026.
- Leo Benac, Jonas Raedler, Zilin Ma, and Finale Doshi-Velez. A benchmark for multi-party negotiation games from real negotiation data, 2026.
- David Carmel and Shaul Markovitch. Opponent modeling in multi-agent systems. In *Adaptation and Learning in Multi-Agent Systems*, volume 1042 of *Lecture Notes in Computer Science*, pp. 40–52. Springer, 1996. doi: 10.1007/3-540-60923-7_18.
- Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail? In *Advances in Neural Information Processing Systems*, 2025.

- Jiaju Chen, Bo Sun, Yuxuan Lu, Yun Wang, Dakuo Wang, and Bingsheng Yao. CollabSim: A CSCW-grounded methodology for investigating collaborative competence of LLM agents through controlled multi-agent experiments, 2026.
- Jiayu Chen, Zelai Xu, Yunfei Li, Chao Yu, Jiaming Song, Huazhong Yang, Fei Fang, Yu Wang, and Yi Wu. Accelerate multi-agent reinforcement learning in zero-sum games with subgame curriculum learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(10):11320–11328, 2024. doi: 10.1609/aaai.v38i10.29011.
- Logan Cross, Violet Xiang, Agam Bhatia, Daniel L. K. Yamins, and Nick Haber. Hypothetical minds: Scaffolding theory of mind for multi-agent tasks with large language models. In *International Conference on Learning Representations*, 2025.
- Peilin Feng, Suorong Yang, and Soujanya Poria. Σ -Mem: An online reliability memory for LLM-based multi-agent systems, 2026.
- Jakob Foerster, Francis Song, Edward Hughes, Neil Burch, Iain Dunning, Shimon Whiteson, Matthew Botvinick, and Michael Bowling. Bayesian action decoder for deep multi-agent reinforcement learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 1942–1951. PMLR, 2019.
- Piotr J. Gmytrasiewicz and Prashant Doshi. A framework for sequential planning in multi-agent settings. *Journal of Artificial Intelligence Research*, 24:49–79, 2005. doi: 10.1613/JAIR.1579.
- Harsh Goel, Aditya Sai Ellendula, Vaishnav Tadiparthi, Ehsan Moradi Pari, Hossein Nourkhiz Mahjoub, and Sandeep P. Chinchali. Bayesian partner modelling enables adaptive replanning for LLM coordination, 2026.
- Keyu He, Xuhui Zhou, and Maarten Sap. Social gym and SPaRTan: Benchmarking and improving LLM social reasoning via multi-agent game tournaments, 2026a.
- Yidong He, Yutao Lai, Pengxu Yang, Jiarui Gan, Jiexin Wang, Yi Cai, and Mengchen Zhao. Strat-reasoner: Reinforcing strategic reasoning of LLMs in multi-agent games. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026b.
- Hengyuan Hu, Adam Lerer, Alex Peysakhovich, and Jakob Foerster. Other-Play for zero-shot coordination. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pp. 4399–4410. PMLR, 2020.
- Wenyue Hua, Zachary Huang, Tyler Payne, Safoora Yousefi, Saleema Amershi, and Asli Celikyilmaz. From passive delegates to strategic negotiators: Reinforcing social reasoning in small language models with SocialRL, 2026.
- Yizhe Huang, Anji Liu, Fanqi Kong, Yaodong Yang, Song-Chun Zhu, and Xue Feng. Efficient adaptation in mixed-motive environments via hierarchical opponent modeling and planning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 20004–20022. PMLR, 2024.
- EunJeong Hwang, Yuwei Yin, Giuseppe Carenini, Peter West, and Vered Shwartz. Infusing theory of mind into socially intelligent LLM agents. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 11327–11360. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.551.
- Kunal Jha, Wilka Carvalho, Yancheng Liang, Simon Shaolei Du, Max Kleiman-Weiner, and Natasha Jaques. Cross-environment cooperation enables zero-shot multi-agent coordination. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 27198–27220. PMLR, 2025.
- Kunal Jha, Aydan Huang, Eric Ye, Natasha Jaques, and Max Kleiman-Weiner. Modeling others’ minds as code. In *International Conference on Learning Representations*, 2026.

- Huaoli Li, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Michael Lewis, and Katia Sycara. Theory of mind for multi-agent collaboration via large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 180–192. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.13.
- Yuxuan Li, Aoi Naito, and Hirokazu Shirado. Systematic failures in collective reasoning under distributed information in multi-agent LLMs. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026a.
- Zeyuan Li, Lukas Petersson, Alessandro Acquisti, and Michiel A. Bakker. Emergent misaligned communication in long-horizon multi-agent LLM commerce, 2026b.
- Jonathan Light, Min Cai, Sheng Shen, and Ziniu Hu. AvalonBench: Evaluating LLMs playing the game of avalon, 2023.
- Bo Liu, Leon Guertler, Simon Yu, Zichen Liu, Penghui Qi, Daniel Balcells, Mickel Liu, Cheston Tan, Weiyan Shi, Min Lin, Wee Sun Lee, and Natasha Jaques. SPIRAL: Self-play on zero-sum games incentivizes reasoning via multi-agent multi-turn reinforcement learning. In *International Conference on Learning Representations*, 2026.
- Nuoyan Lyu, Bingbing Xu, Weihao Meng, Yige Yuan, Yang Zhang, Zhiyong Huang, Tat-Seng Chua, and Huawei Shen. GIFT: Games as informal training for generalizable LLMs, 2026.
- Owen Macindoe, Leslie Pack Kaelbling, and Tomás Lozano-Pérez. POMCoP: Belief space planning for sidekicks in cooperative games. *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, 8(1):38–43, 2012. doi: 10.1609/aiide.v8i1.12510.
- Hachem Madmoun and Salem Lahlou. Communication enables cooperation in LLM agents: A comparison with curriculum-based approaches. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 307–321. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.eacl-short.23.
- Alexander Meulemans, Maciej Wołczyk, Marissa A. Weis, Rajai Nasser, Roberta Rocca, Seijin Kobayashi, Guillaume Lajoie, Angelika Steger, Blake Richards, Marcus Hutter, James Manyika, Rif A. Saurous, João Sacramento, and Blaise Agüera y Arcas. A game theory for foundation models shows new paths to rational cooperation through similarity inference, 2026.
- Chunjiang Mu, Ya Zeng, Qiaosheng Zhang, Kun Shao, Chen Chu, Hao Guo, Danyang Jia, Zhen Wang, and Shuyue Hu. Adaptive theory of mind for LLM-based multi-agent coordination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 29608–29616, 2026. doi: 10.1609/aaai.v40i35.40204.
- Abigail O’Neill, Alan Zhu, Mihran Miroyan, Narges Norouzi, and Joseph E. Gonzalez. Cooperate to compete: Strategic coordination in multi-agent conquest. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026.
- Matthew Riemer, Zahra Ashktorab, Djallel Bouneffouf, Payel Das, Miao Liu, Justin D. Weisz, and Murray Campbell. Position: Theory of mind benchmarks are broken for large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 82091–82130. PMLR, 2025.
- Jan Sobotka, Mustafa O. Karabag, and Ufuk Topcu. Why do LLMs struggle in strategic play? broken links between observations, beliefs, and actions, 2026.
- Joseph West, Frederic Maire, Cameron Browne, and Simon Denman. Improved reinforcement learning with curriculum. *Expert Systems with Applications*, 158:113515, 2020. doi: 10.1016/j.eswa.2020.113515.
- Jincenzi Wu, Yuxuan Lei, Jianxun Lian, Yitian Huang, Lexin Zhou, Haotian Li, Xing Xie, and Helen Meng. Social-R1: Towards human-like social reasoning in LLMs, 2026.

- Sarah A. Wu, Rose E. Wang, James A. Evans, Joshua B. Tenenbaum, David C. Parkes, and Max Kleiman-Weiner. Too many cooks: Bayesian inference for coordinating multi-agent collaboration. *Topics in Cognitive Science*, 13(2):414–432, 2021. doi: 10.1111/tops.12525.
- Tim Wyse, Kaitlin Bustos, Yulia Volkova, and Max Kleiman-Weiner. Commitment to cooperation with self-negotiated contracts. In *Proceedings of the Third Conference on Language Modeling (COLM)*, 2026.
- Keyang Xuan, Pengda Wang, Chongrui Ye, Haofei Yu, Tal August, and Jiaxuan You. SocialVeil: Probing social intelligence of language agents under communication barriers. In *Conference on Language Modeling (COLM)*, 2026.
- Yiheng Yao, Chelsea Zou, and Robert D. Hawkins. Talk is cheap, communication is hard: Dynamic grounding failures and repair in multi-agent negotiation, 2026. Accepted at the Conference on Language Modeling (COLM 2026).
- Yashwanth YS, Ruichen Wang, Shihua Zeng, Xuhui Zhou, Koichi Onoue, Vasudha Varadarajan, and Maarten Sap. SOTOPIA-TOM: Evaluating privacy and information management in multi-agent interaction with theory of mind. In *Conference on Language Modeling (COLM)*, 2026.
- Huining Yuan, Zelai Xu, Zheyue Tan, Xiangmin Yi, Mo Guang, Kaiwen Long, Haojia Hui, Boxun Li, Xinlei Chen, Bo Zhao, Xiao-Ping Zhang, Chao Yu, and Yu Wang. MARSHAL: Incentivizing multi-agent reasoning via self-play with strategic LLMs. In *International Conference on Learning Representations*, 2026.
- Erica Zhang, Fangzhao Zhang, Aneesh Pappu, Batu El, Jose Blanchet, Susan Athey, Jiashuo Liu, and James Zou. TERMS-Bench: Diagnosing LLM negotiation agents beyond deal rate, 2026a.
- Yuzhe Zhang, Feiran Liu, Yi Shan, Xinyi Huang, Xin Yang, Yueqi Zhu, Xuxin Cheng, Cao Liu, Ke Zeng, Terry Jingchen Zhang, and Wenyuan Jiang. SILO-BENCH: A scalable environment for evaluating distributed coordination in multi-agent LLM systems. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 29379–29398. Association for Computational Linguistics, 2026b. doi: 10.18653/v1/2026.acl-long.1354.
- Ruiwen Zhou, Maojia Song, Xiaobao Wu, Sitao Cheng, Xunjian Yin, Yuxi Xie, Zhuoqun Hao, Wenyue Hua, Liangming Pan, Soujanya Poria, and Min-Yen Kan. Epistemic context learning: Building trust the right way in LLM-based multi-agent systems, 2026.
- Kunlun Zhu, Hongyi Du, Zhaochen Hong, Xiaocheng Yang, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Xiangru Tang, Heng Ji, and Jiaxuan You. MultiAgentBench: Evaluating the collaboration and competition of LLM agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8580–8622. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.421.
- Chelsea Zou, Yiheng Yao, Selena She, Noah D. Goodman, and Robert D. Hawkins. CalBench: Evaluating coordination-privacy trade-offs in multi-agent LLMs, 2026.

A NATIVE-SEMANTIC DIAGNOSTIC PROTOCOL

Frozen matched structures. The diagnostic contains 16 held-out structures, eight with binary goals and eight with linear goals. Each structure contributes a voluntary and a preset case. The two cases have byte-identical current physical states, legal-action sets, per-world payoff tables, and fixed continuation policies. They differ only in whether the displayed prefix was chosen by the reference partner or externally imposed. Model outputs were not used for selection. CPU certificates verify that the two cases have different correct semantic judgments and disjoint invariant optimal-action sets. The exact-posterior gap between the best and best non-optimal action ranges from 0.33 to 1.00 terminal-utility units (mean 0.67), ruling out nominal differences caused only by numerical ties.

The queried uncertainty is one partner-goal preference in {WANT, NEUTRAL, AVOID}. The deployed *B* interface returns the complete supported subset and either its uniquely favored member

or UNDETERMINED; it does not return probabilities. Both the model and correct judgments use identical serialization. The B prompt necessarily identifies whether an event was observed or imposed; the controlled P prompt contains neither that evidence nor its provenance. Condition names, case identifiers, correctness claims, posterior probabilities, and payoff values are absent from model prompts. Exact posteriors remain in the scorer and are used with the common per-world payoff table to compute Eq. 4.

The correct semantic judgment is not a complete posterior, so selection explicitly certifies that the omitted magnitudes are decision-irrelevant. For each case, we express its support and favored-margin semantics as a finite union of rational convex polytopes and enumerate every vertex exactly. We also include zero-mass and threshold boundaries, obtaining a conservative superset of the strict semantic region. All vertices must induce the same exact optimal-action set. Since expected action value is affine in the posterior, this proves invariance over every compatible posterior; all 32 retained cases pass. The invariant voluntary and preset sets must additionally be disjoint.

Calls and interventions. For each case we obtain: (i) a native semantic B judgment; (ii) a fresh P call supplied with that judgment; (iii) a fresh P call supplied with B^* ; and (iv) a native history-to-action call with no explicit judgment. The two controlled P calls include the current binding state, remaining turns, investigation quota, known preferences, legal actions, and fixed partner policy, but exclude the behavioral chronology and its voluntary/preset provenance. Their tools, sampling parameters, and request seed are identical; only the serialized judgment may differ. Moreover, after removing that judgment, the controlled- P request is byte-identical across the two cases in each matched structure. If the model judgment is invalid, its composition cell is missing rather than replaced by a default action. On eight voluntary cases, a fifth call supplies B with the reference planner’s qualitative likelihood ordering while still withholding all numeric probabilities.

Selection and scoring use the fixed teacher only. In particular, changing a judgment never recomputes the partner continuation policy, and per-world payoffs respect the focal player’s information set rather than optimizing a different continuation separately in each hidden world. The paired quantity Δ_B is therefore the effect of changing the semantic input to the same model planner. It can have either sign and is not combined with planning regret as an additive error decomposition. Condition-specific regret means may have different valid subsets. We therefore also aggregate both regrets and Δ_B on the identical doubly valid case/replica pairs and identical structure weights, where the implementation checks $\Delta_B = \text{Reg}(\hat{B}) - \text{Reg}(B^*)$ row by row and after aggregation.

Execution and determinism. We evaluate Qwen3-4B-Instruct-2507 through vLLM and the Hermes tool-call parser, using greedy decoding ($T = 0$, $\text{top-}p = 1$, $\text{top-}k = -1$), a 4,096-token completion limit, and three replicas. The vLLM workers enable batch-invariant execution; repeated requests must agree even when production concurrency and chunked prefill are retained. We additionally require that whenever $\hat{B} = B^*$, the two controlled- P request bodies (including their seed) are identical and return the same action. Replicas are determinism checks, not independent statistical samples. Both checkpoint runs contain 96/96 rows and 408/408 calls. All repeated outputs agree; all semantic certificates pass; and the paired-scoring identity has zero row failures (maximum residual 0).

Aggregation. We first collapse the identical replicas, average valid cases within each structure, then weight structures equally. The old and new runs have 16 and 24 doubly valid paired cases spanning 11 and 14 structures. Their paired repair means use those respective subsets. For checkpoint comparison we additionally intersect validity masks: 14 paired cases across 10 structures yield repair -0.158 (old) and -0.550 (new); 24 correct- B cases across all 16 structures yield regret 0.802 and 1.130; and 28 end-to-end cases across 14 structures yield 0.690 and 0.887. No mean is reconstructed by subtracting metrics computed on different valid subsets.

The frozen v6 suite stores the model-facing cases, teacher-only certificates, selection record, and both clean controlled- P and native end-to-end requests. CPU checks reject any controlled planning prompt containing a behavioral chronology or provenance marker and verify byte identity across each matched pair before judgment insertion. Hashes in the suite manifest detect changes to frozen cases and label dependencies before inference.